

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:BE02000011****Date:18-06-2025****Subject Name:Mathematics – 2****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- | | Marks |
|--|-----------|
| Q.1 (a) Prove that $f(z) = \frac{1}{z}$ is not differentiable at $z = 0$, by definition. | 03 |
| (b) Solve $\frac{dx}{dy} + \frac{x}{y} = x^3$. | 04 |
| (c) Check the whether the given matrix $A = \begin{bmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{bmatrix}$ is diagonalizable?
If so, then find matrix P such that $P^{-1}AP = D$. | 07 |
| Q.2 (a) State Rouché's Theorem. Evaluate $\int_C e^z dz$ if C is the circle $ z = 99$. | 03 |
| (b) Solve $y'' + 4y' + 4y = 0, y(0) = 1, y'(0) = 1$. | 04 |
| (c) (1) Evaluate $\int_C \frac{3z^2+z}{z^2-1} dz$ where C is the circle $ z - 1 = 1$. | 03 |
| (2) Define entire function. Prove that e^z is analytic at everywhere. | 04 |
| OR | |
| (c) (1) Evaluate $\int_C \operatorname{Re}(z) dz$ from $1 + i$ to $3 + 2i$ along the straight line $2y = x + 1$. | 03 |
| (2) Find real part and imaginary part of the following functions
(i) $\sin z$ (ii) $\ln z$. | 04 |
| Q.3 (a) Solve $\frac{dy}{dx} + 2y \tan x = \sin x, y(0) = 2$. | 03 |
| (b) Solve the following linear system by Gauss elimination method:
$x - 2y + 3z = -2, \quad -x + y - 2z = 3, \quad 2x - y + 3z = -7$. | 04 |
| (c) Define Harmonic function. Check whether $v = 3x^2y - y^3$ is harmonic or not. If so, construct the analytic function $f(z) = u + iv$. | 07 |
| OR | |
| Q.3 (a) Test for exactness and solve $2xydx + x^2dy = 0$. | 03 |
| (b) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ -1 & 3 & 0 & -4 \\ 2 & 1 & 3 & -2 \\ 1 & 1 & 1 & -1 \end{bmatrix}$. | 04 |
| (c) Check whether $u = 3x - 2xy$ is harmonic or not. If so, find the harmonic conjugate v of u . | 07 |

- Q.4 (a)** Solve $x^2 y dx - (x^3 + xy^2) dy = 0$. **03**
- (b)** Find the inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ using Gauss Jordan method. **04**
- (c)** Find all the Taylor's and Laurent series of $f(z) = \frac{-2z+3}{z^2-3z+2}$ with center 0. **07**

OR

- Q.4 (a)** Solve $xe^x(dx - dy) + e^x dx + ye^y dy = 0$. **03**
- (b)** Find inverse of the matrix $A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ by Cayley-Hamilton theorem. **04**
- (c)** State Cauchy Residue theorem. Evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$ where C is the circle $|z| = 3$ using Cauchy Residue theorem. **07**
- Q.5 (a)** Solve $x^2 p^2 + 3xyp + 2y^2 = 0$. **03**
- (b)** Solve $xy'' + y' = \frac{12 \log x}{x}$. **04**
- (c)** Solve $y'' - 2y' + y = xe^x \sin x$ by using the method of variation of parameter. **07**

OR

- Q.5 (a)** Solve $\left(x \frac{dy}{dx} - y\right) \left(1 - \frac{dy}{dx}\right) = \frac{dy}{dx}$. **03**
- (b)** Solve $y'' + 2y' + 4y = 2x^2 + 3e^{-x}$ by undetermined coefficient method. **04**
- (c)** Find the power series solution of $y'' + xy' + x^2y = 0$ about $x = 0$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025****Subject Code: BE02000011****Date: 23-12-2025****Subject Name: Mathematics – 2****Time: 02:30 PM TO 05:30 PM****Total Marks: 70****Instructions**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Check whether the function $e^{\bar{z}}$ analytic or not at any point ?	03
(b) Solve $x \log x \frac{dy}{dx} + y = 2 \log x$	04
(c) Diagonalize the matrix $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$	07
Q.2 (a) State Cauchy integral formula and Evaluate $\oint_C \frac{dz}{z(z+3)}$ Where C is the circle $ z = 1$	03
(b) Solve $y'' - 4y = 0, y(0) = 2, y'(0) = -1$	04
(c) (i) Evaluate $\int_C (x - y + ix^2) dz$ Where C is a straight line from $z=0$ to $z=1+i$. (2) If $\sin(\alpha + i\beta) = x + iy$ then prove that $x^2 \operatorname{sech}^2 \beta + y^2 \operatorname{cosech}^2 \beta = 1$	07
OR	
(c) (i) Evaluate $\oint_C \frac{e^z}{(z-1)^4} dz$ where C is the circle $ z = 3$ (ii) Using CR equations show that $z z $ is no where analytic.	07
Q.3 (a) Solve $\frac{dy}{dx} + 2xy = xe^{-x^2}$	03
(b) Find the rank of the matrix $A = \begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$	04

- (c) Check whether $u = x^2 - y^2 - y$ is harmonic or not. If so find the harmonic conjugate v of u. 07

OR

- (a) Test for exactness and solve $(\tan y + x) dx + (x \sec^2 y - 3y) dy = 0$ 03

- (b) Solve the following linear system by Gauss elimination method. 04
 $x + 2y - z = 2, x - y + z = 4, 2x + y - z = 5$

- (c) Define Harmonic function. Check whether $u = x^3 - 3xy^2$ is Harmonic or not if so construct the analytic function $f(z) = u + iv$ 07

- Q.4 (a) Solve $(xy^2 - x) dx = (y + x^2y) dy$ 03

- (b) 04

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

 Find inverse of the matrix by Cayley Hamilton theorem.

- (c) 07
 State Cauchy residue theorem and Evaluate $\oint_C \frac{z^3 - z^2 + z - 1}{z^3 + 4z} dz$, Where C is
 (a) $|z| = 1$, (b) $|z| = 3$

OR

- (a) 03
 Solve $x \frac{dy}{dx} + \frac{y^2}{x} = y$

- (b) 04

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

 Find the inverse of the matrix using Gauss Jordan method.

- (c) 07
 Expand $f(z) = \frac{1}{(z+1)(z-2)}$ in Laurent's series in the region
 (1) $|z| < 1$ (2) $1 < |z| < 2$ (3) $|z| > 2$

- Q.5 (a) Solve $(x^2 + y^2 + x) dx + xy dy = 0$ 03

- (b) 04
 Solve $y'' - 2y' + y = 4 \frac{e^x}{x^2}$

- (c) 07
 Solve $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$ by the method of variation of Parameters.

OR

- (a) Solve $y = 3x + \log p$ 03

- (b) 04
 Solve $y'' + y' - 6y = 6x + 3x^2 - 6x^3$ by undetermined coefficient method

- (c) 07
 Find the power series solution of $y'' + x^2y = 0$ about $x = 0$.
